# Report : ATMEGA32A based kit

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## Exploring GPIO

Learnt about GPIO and the registers associated with it.

DDR* - Data Direction Register used for setting input/output fundamentally.

PORT* - Sets Output Level and in case of a input used to set the internal pullups.

PIN* - Used to read the logic level at the pin.

* - A,B,C,D (for ATMEGA32A)

Made a simple blink example where i used a builtin delay function.
See the folders blink and Input for examples associated with GPIO.

## Compiling and Flashing

Used avr-gcc and avr-objcopy for compiling and linking.
Used avrdude for flashing.

**avr-gcc -mmcu=atmega32a -DF_CPU=8000000UL  -Os -Wall -o blink.elf blink.c**

Here:

• -mmcu=atmega32a selects the target microcontroller.

• -DF_CPU=8000000UL defines the CPU clock frequency for the source code.

• -Os enables optimizations for code size.

• -Wall enable additional compiler warnings.

• -o blink.elf specifies the output executable.

**avr-objcopy -O ihex -R .eeprom blink.elf blink.hex**

**avrdude -p m32 -c usbisp -U flash:w:blink.hex:i**

Learnt how to create a makefile and how to use make.
Make makes the compiling and configuration steps very much simpler.

# Timers and PWM

Learnt how to configure a timer in avr-c and registers related to timer.

Atmega32A has three timers, where Timer0 and Timer2 are 8 bits and Timer1 is 16 bits.

TCCR* - Timer Configuration and Control Register.

TCNT* - Timer Current Value present at this register.

OCR* - Output Compare Register used for setting duty cycle in pwm mode.

TIMSK – Used for enabling Timer interupts

Timer 1 has two compare registers OCR1A and OCR1B used for accomordating 16bit values.

Generated a PWM signal and viewed in a Digital Ocilloscope.

Created a program where we can change the duty cycle and frequency of the gererating PWM on the go using Button inputs.

See the folder timer for the above example.

# I2C

Learnt about I2C protocol and how to configure it in avr-c.
I2C has conditions like Start, Stop, ACK and NACK.

TWBR – Bitrate Register used to configure bitrate for I2C.
TWSR – Status Register contains flags for Ack.
TWDR – Data Register used to send and revcieve data using I2C
TWCR – Control Register used to generate start and stop conditions and initiating a transaction.

Learnt about interfacing I2C based devices like DS1307 (RTC) and 2404C (EEPROM).
Learnt about the device's memory structure( Both MCU and I2C Devices).

Learnt about bitbanging and tried to bitbang I2C and accessed the I2C eeprom ic (2404C).

I2C works on a open drain concept. The entire bus is pulled up and the device can only pull the SDA and SCL lines low.
Creating a alternative for HIGH and LOW, While bit banging we use Output LOW for LOW and for HIGH we change the DDR* to input so that the pin goes to the tristate releasing the bus.

**HIGH –> RELEASE** in open drain output.

We need to just create basic functions to emulate start , stop and transmit with/without ack.
Then we can easily replace the existing hardware functions with it.

Learnt about the limitations of the Atmega32 while bitbanging as it cannot able to achieve standard 300kHz I2C clock with bitbanging.

See the folders rtc, eeprom, eeprom_bb for examples.

# UART

Learnt about UART protocol and how to configure it in avr-c

Common UART Configuration is 8N1 – 8 bits , No Parity, 1 Stop Bit

UBRRH,UBRRL – USART Baud Rate Registers. Value Calculated based on the Baudrate value.

UCSRA – Contains Status flags. Like Register Empty.
UCSRB – Uesd for enabling UART

UCSRC – Used for setting frame (8N1)

UDR – Data Register for transmission

Created example programs to send data to PC via UART.
Used screen command in Linux to view the transmitted data.

**screen /dev/ttyUSB0 9600**

Opens the serial port ttyUSB0 with baudrate 9600.

See the folders matrix_uart and temp_uart for examples.

# ADC

Learnt about ADC process , form sampling to qauntization.
Learnt how to configure ADC in avr-c.

ADC can be done in two methods , single ended and differential.
ADC take a time to convert the analog to digital.
Higher Gains Higher Convertion Time.

ADMUX – Used for selecting the ADC channel.
ADCSR – Control and Status Register used for starting a conversion and configuration.

ADCH,ADCL – Contains the Converted Data (10 bits so two registers)

SFIOR – Used for enabling auto trigger mechanism based on timer.

Created a example where the temperature is sensed from a LM35 sensor via ADC conversion.

# 16x2 LCD Display

Learnt about this display from its datasheet.
Learnt about the command and the memory structure of it.
Used it in some examples like rtc for output.
Explored the ablities to shift the display left/right.

The display has CGROM that has predefined set for characters mostly matched by ascii value.
The CGRAM has 64 Bytes used to create custom characters.
The DDRAM contains the 40x2 dimension where any 16x2 section can be displayed on the display by shifting left/right.

The trainsmission is done by setting the appropraite logic at R/W pin and setting/reading the Data bus. We need to pulse the enable pin for the transaction

There are many command available for accessing the lcd that need to be provided on the data bus. Some of them are Shifting Display, Writing to DDRAM/CGRAM, Changing Cursor Position.

# Matrix Kepad

Learnt how to multiplex and read , reducing the no of pins.

The keypad will be in row x col configuration, where either row/col pins are used for reading/scanning alternatively.
These usually work on Logic HIGH. The scan lines are multiplixed and the read lines are read on each cycle of multiplexing.

# DS1307

It is a I2C based RTC IC that can do timekeeping.
It has seperate registers to store seconds, minutes, hours, date, month & year.
All values present in BCD format. In BCD format individual digits stored as  a 4bit value.
Additionally it has 56 bytes RAM that is battery backed similar to other registers.

# Reference

All datasheets are under the datasheet folder for reference.
There codes are present in seperate folders with the makefile.